

Troubleshooting biGBac: a practical guide

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Abstract

The biGBac system presented by Weissmann *et al.* (Weissmann *et al.*, 2016 and Weissmann *et al.*, 2018) is a modular insect cell expression system designed for expression of multi-subunit complexes. The system is very powerful, allowing up to 25 genes to be assembled into one baculo-virus expression vector, but does require a reasonable amount of cloning experience in order to attain clones in the 6 day time frame stated in the paper. This guide was written to provide a detailed explanation how the system is designed, how it can be further exploited and provide in-depth practical information that even a novice cloner can follow.

Introduction

The Design of the System

The biGBac system uses the Gibson assembly method, where homologous double-stranded regions and a cocktail of enzymes are used to join multiple DNA fragments together (Gibson *et al.*, 2009). There are many online guides to how this works and many different versions of Gibson assembly enzyme mix available commercially. In brief, the assembly mix contains an exonuclease, a polymerase and a ligase. On incubating the DNA fragments with assembly mix at 50°C, the exonuclease chews the DNA back 5' to 3', leaving complementary single stranded regions, that can anneal. This leaves gaps in the DNA backbone, which are filled by the polymerase and ligase, resulting in a seamless vector (Figure 1). The biGBac system uses this method to assemble multiple genes together into vectors.

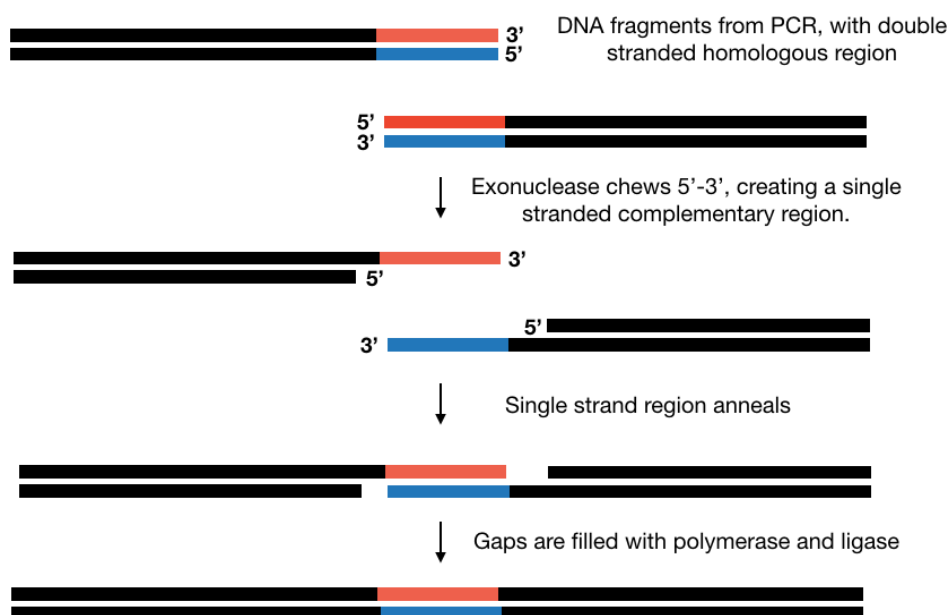


Figure 1: The steps of Gibson assembly cloning.

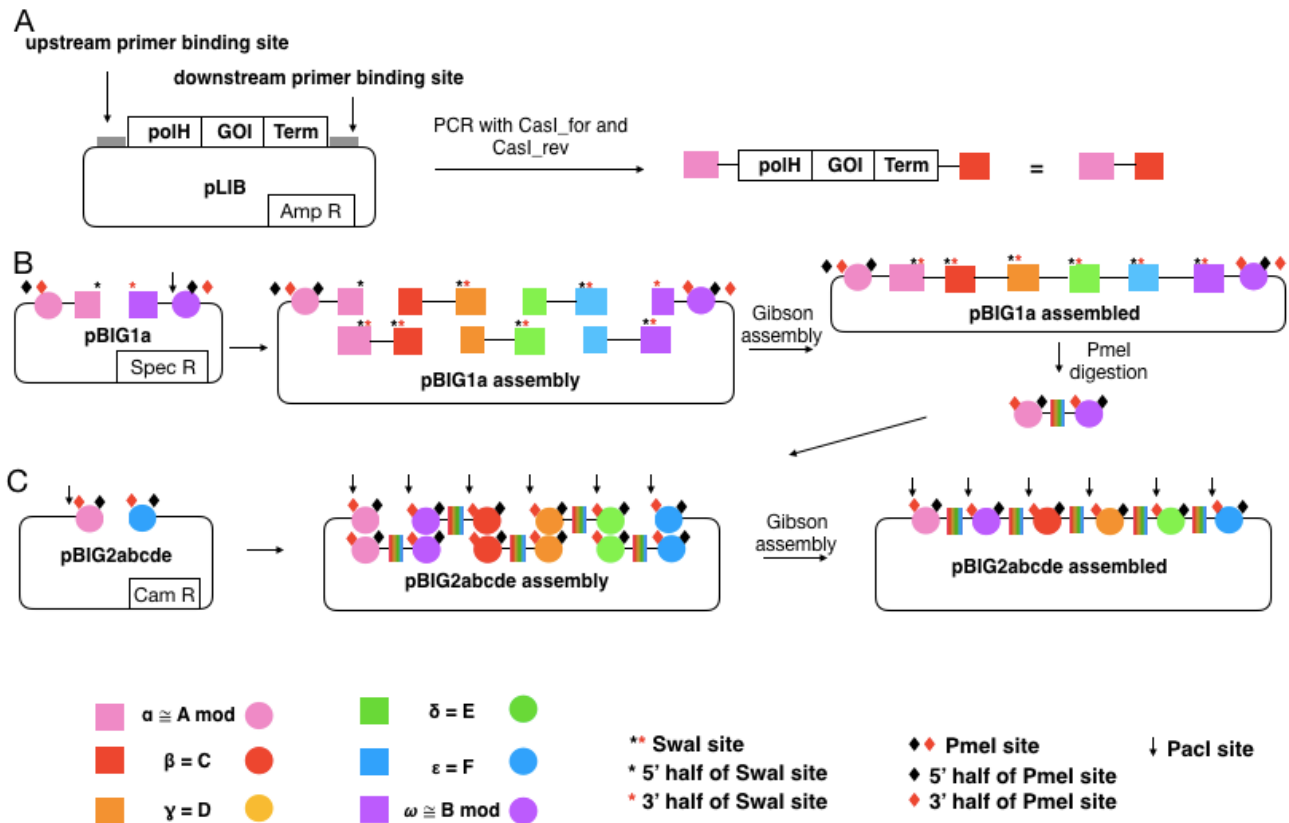


Figure 2: The vectors of the biGBac system. A: pLIB vectors contain one gene of interest (GOI) cloned under the control of a polH promoter and terminator to make a gene expression cassette (GEC). B: pBIG1 vectors can store up to 5 GECs, which are assembled by Gibson assembly with overlapping regions added by PCR. C: pBIG2 vectors can store up to 25 GECs, assembled from up to 5 pBIG1 vectors, with overlapping regions provided in the vectors.

There are three different levels of vectors in the biGBac system; pLIB, pBIG1s and pBIG2s (Figure 2, Table 1). The pLIB vectors store one gene of interest (GOI), adding a polyhedrin promoter and terminator to generate a gene expression cassette (GEC). The pBIG1 vectors hold up to 5 GECs, which are assembled from PCR fragments with a set of overlapping regions labelled α , β , γ , δ , ϵ and ω using Gibson assembly. The pBIG2 vectors which hold up to 25 GECs, and are made by up to five pBIG1 restriction fragments which are assembled using the overlapping regions labelled A, B, C, D, E and F. All the vectors have a different antibiotic resistance gene, to help reduce background. All the vectors can also be transposed into DH10embacY cells for bacmid production, (hence suitable for protein production in insect cells), so I encourage you to exploit this and make pBIG1 constructs that should form useful, stable sub-complexes.

To assemble pBIG1 vectors, the vector is linearised using digestion with the *SwaI* enzyme, leaving blunt ends, with one end having the α assembly overlap region and the other end having the β region. PCR is then performed using a different set of primers for each different pLIB GOI. Each primer has a region that is homologous to the pLIB vector and a region that adds to double stranded overlap for assembly; α , β , γ , δ , ϵ or ω (Figure 3). In addition, all reverse primers, and the first forward primer, **CasI_for**, have a *SwaI* restriction enzyme site, hence the assembled vector should have a *SwaI* site between every GEC such

that the finally assembly can be screened using diagnostic digest with *Swa*I. Primers pairs are chosen to ensure assembly overlaps are made between adjacent genes and the vector, for example the first gene DNA fragment has an α assembly overlap added to the 5' to overlap with the vector, and a β to the 3'. The second gene has a β overlap added to the 5' to overlap with the first gene and a γ to the 3' to overlap with the next gene. The final gene has a ω at the 3' to overlap with the vector (Figure 2). In addition, pBIG1 vectors each have another set of overlapping regions on either side of the GECs, labelled A, B, C, D, E or F. These have flanking *Pme*I sites, such that digesting pBIG1 vectors with *Pme*I produces a DNA fragment with A, B, C, D, E or F on the 5' and 3' ends, which in turn allow assembly overlaps with the pBIG2 vectors.

Depending on which pBIG1 vector you used, this digest product will have the regions, A, B, C, D, E or F on the end. All pBIG2 vectors have the overlap region A, but vary in the second region. For example, pBIG2ab has overlap A and C, hence you can assemble a fragment from pBIG1a and 1b. Vector pBIG2abc has overlap A and D, hence you can assemble pBIG1a, 1b and 1c. In the pBIG2 vectors, and in pBIG1 vectors between the ω and the B, C, D, E or F regions, there is a *Pac*I restriction site, hence pBIG2 assembly can be checked with diagnostic digestion with *Pac*I.

Hence, the system can just be thought of as a set of Gibson assembly homology regions that allow DNA fragment assembly and contain restriction enzyme sites to release DNA fragments and screen for correct assemblies.

	Variable region α forward	<i>Swa</i> I site	pLIB homology region
CasI_for	AACGCTCTATGGTCTAAAG	ATTAAAT	CGACCTACTCCGGAATATTAATAGATC
	Variable region β reverse		
CasI_rev	AAACGTGCAATAGTATCCAGTT	ATTAAAT	GGTTATGATAGTTATTGCTCAGCG
	Variable region β forward		
CasII_for	AAACTGGATACTATTGCACGTTT	AAAT	CGACCTACTCCGGAATATTAATAGATC
	Variable region γ reverse		
CasII_rev	AAACATCAGGCATCATTAGGTT	ATTAAAT	GGTTATGATAGTTATTGCTCAGCG
	Variable region γ forward		
CasIII_for	AAACCTAATGATGCCTGATGTTT	AAAT	CGACCTACTCCGGAATATTAATAGATC
...			
	Variable region ω reverse		
Casω_rev	AACCCCGATTGAGATATAGATT	ATTAAAT	GGTTATGATAGTTATTGCTCAGCG

Figure 3: The structure of the biGBac primers. The blue regions are homologous to the region outside the GEC of the pLIB vector and are the regions that anneal to the pLIB vectors in PCR. The red part of the sequence are the Gibson assembly regions that are added to the GEC during PCR. *Swa*I sites are also added using all the reverse primers and the first forward primer.

Practical Aspects of the biGBac System

Cloning in this system is not always straightforward, as the probability of multiple DNA fragments assembling correctly goes down the more fragments you try to assemble together. This can go wrong substantially in two ways: 1) formation of partially assembled products, where one or two genes gets assembled, but not all of them, or 2) formation of empty vectors. In our experience three gene assemble well, assembling four will require a lot of screening, and I wouldn't recommend trying to assemble 5 genes in one go if you can avoid it. All genes will not behave the same in the system, for example it can be harder to clone large (>8kb) genes together, especially with other genes of very different size. In practice, using diagnostic digest with *SwaI* and *PacI* can be very time consuming, as it requires an overnight culture. If you don't pick the right colony for your overnight culture, you have to wait to have another go. Some members in the lab have had to screen more than 40 colonies to find a fully assembled clone, wasting a lot of time and reagents. To get over this, we have introduced another screening step using colony PCR. This uses four primers (Table 2, primers S1A-D for pBIG1s and S2A-D for pBIG2s), where two primers bind the vector and two primers bind each fragments. These primers will produce a different sized PCR product across the vector-fragment, fragment-fragment and fragment-vector junctions; hence colonies where two or more genes have been assembled will have three bands (Figure 4). Colonies with empty vector, however will only yield one band. The primers are different for pBIG1 and pBIG2 assemblies but in both work using the same principal.

Colony PCR is a very quick reaction and makes it possible to screen many more colonies, without having to wait for an overnight culture, however if you have three or more genes being assembled, it may not be able to discriminate between two or three (or more) genes being assembled. Hence, in practice, screening with both colony PCR and diagnostic digest gives the highest chance of success.

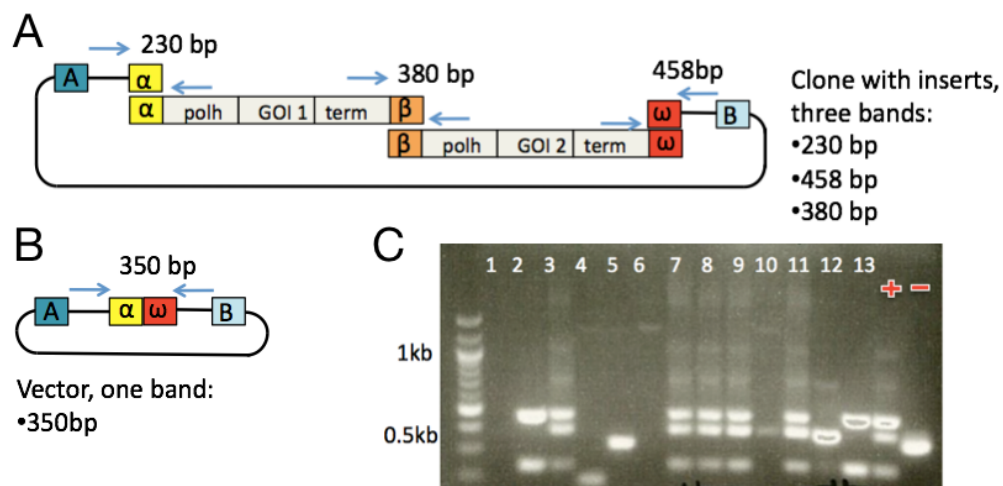


Figure 4: Colony PCR for screening pBIG1 constructs. A: Two primers bind the pBIG1 vector, facing towards the fragments, two primers bind the fragments, facing the vector. This results in three products across the assembly junctions, of 230 bp, 380 bp and 458bp, corresponding to vector-fragment 1, fragment 1-fragment 2 (and fragment 2-fragment 3, etc) and the last fragment -vector. B: The empty vector will only product one PCR product, of ~350 bp. C: Example of colony PCR 2% agarose gel; the red + is a positive control pBIG1 from a previous cloning, red - is the pBIG1a empty vector and colony 7, 8, 9 and 11 proved be correctly assembled.

A feature that is built in this system that isn't immediately obvious, is that the Cas primers' Gibson assembly overlaps to assemble the pBIG1 vectors are actually the same as to assemble the pBIG2. That is, overlapping regions $\beta=C$, $\gamma=D$, $\delta=E$ and $\epsilon=F$. This can be useful if you want to avoid making a pBIG1 with just one gene in it. For example, imagine if you had complex made of a large scaffold protein, in addition to a stable heterodimer and a heterotrimer (6 subunits in total). It would make sense to have the heterodimer in pBIG1a and the heterotrimer in pBIG1b, so you can express these subcomplexes separately. You could place the scaffold protein in pBIG1c in order to make a pBIG2abc construct with the scaffold protein, heterodimer and heterotrimer, but you don't have to. Instead you can add the C and D Gibson assembly regions with PCR using **CasII_for** and **CasIII_rev**. This will remove one PaeI screening site, but diagnostic digest with SmaI and PaeI should still be able to identify a unique restriction pattern.

All the pBIG2 vectors start with the A assembly overlap, so what do you do if you want to make combinations of subcomplexes that lack pBIG1a? That is, using the above example, what would I do if I wanted to make a construct that had the scaffold and the heterodimer (combine pBIG1a and pBIG1c) and another construct that had the scaffold and the heterotrimer (combine pBIG1b and pBIG1c)? You can make these constructs by digesting the empty pBIG1 vectors with PmeI and use the product as a linker. For example, to combine pBIG1a and pBIG1c, simply digest pBIG1b, and you will have a 400 bp linker with the B overlap at the 5' end and the C overlap at the 3' end. This is implied by Weissmann *et al* in supplementary Table 4, which states that some pBIG2 constructs contain empty pBIG1 vectors.

Method

Constructs

First you need the biGBac vectors, they are available on addgene:

<https://www.addgene.org/kits/peters-bigbac/>

Next you need to logically design your constructs. As said above, put stable sub-complexes in individual pBIG1 vectors and try not to put more than 3 or 4 genes per pBIG1 vector. I computationally build all constructs I want to using the program APE (A Plasmid Editor):

<http://biologylabs.utah.edu/jorgensen/wayned/ape/>

After building them, I perform computational digests with SmaI and PaeI, such that I can ensure two products won't run at the same size in both digests.

I have all constructs ordered or subcloned lacking restriction enzyme sites used for subsequent steps of biGBac cloning, ie PmeI, SmaI, PaeI, and any other restriction sites for downstream cloning.

pBIG1 Assembly:

PCR each pLIB expression cassette (a promoter, gene and terminator) using the Cas primers in primer Table 2. Each Cas primer will add a sequence (α , β , γ ,

etc) to the 5' and 3' end of the gene expression cassette, which can be used for assembly. The first gene should use **CasI_for**, and **CasI_rev**, the second **CasII_for** and **CasII_rev** and the final, using **Casω_rev**, with the forward version of the second to last reverse primer. I use primers from Integrated DNA Technology with standard de-salting purification (Weissmann used PAGE purified oligonucleotides).

PCR of pLIBs

DNA template mini-prep(100-200ng)	0.5µL
Phusion 5x buffer	10µL
dNTPs (10mM)	1µL
Forward primer (10µM)	2.5µL
Reverse primer (10µM)	2.5µL
Phusion polymerase	1µL
Water	up to 50µL

- Use the phusion cycle with an annealing temperature of 65°C, an extension time of 15-30s per kb of longest gene to be amplified and 30-35 cycles.
- After PCR has finished, perform a gel extraction. (If you are sub-cloning with Gibson assembly, prior to running gel extraction, add 1µL of DpnI enzyme and incubate at 37°C for an hour, this will remove plasmid template.)
- You can also use this protocol to amplify the pBIG1 vectors using the primers **alpha_rev** and **omega_for** from primer table 2, but in this case a DpnI digest is essential.

Digest Vector

While the PCR is going on, digest the pBIG1 vector (s) to be used with Swal.

Mini prep DNA (10µg)	20-30µL
10x buffer 3.1	4 µL
Swal enzyme	1-2µL
Water	Up to 40µL

- Incubate overnight at room temperature (incubation at higher temperatures inactivates Swal). In the morning add a further 1µL of enzyme, and incubate for a further hour before inactivating at 65°C for 20 minutes. It is essential the vectors are well cut, or you will have a high background.
- Gel-extract vector and PCR product in a 1% agarose gel. I would recommend running uncut pBIG1 vector on the same gel, such that you can see the band has shifted completely on digestion. Use the zymo gel extraction kit to purify DNA, as this kit results in very clean and concentrated DNA, which will help the Gibson assembly reaction proceed efficiently. DNA fragment for purification is greater than 8kb, adding equal gel volume of water will improve DNA yield. Elute in 10uL of water and quantify DNA concentration using nanodrop.

Zymo kit: <https://www.bioscience.co.uk/search/132097/>

Assembly Reaction

I had used both the following kits successfully:

<https://www.neb.com/products/e5510-gibson-assembly-cloning-kit#Product%20Information>

<https://www.neb.com/products/e5520-nebuilder-hifi-dna-assembly-cloning-kit#Product%20Information>

You can make your own like Weissmann *et al*, but the kits are a nice entry-level way of cloning with Gibson assembly.

Although the kit recommends using 10µL of 2x assembly mix with 10µL of DNA mix, I find 5µL of each works fine. The first time you use the kit, defrost it slowly on ice, spin down condensation, make 5µL aliquots on ice and freeze until required. I use PCR tubes and use a PCR machine for the assembly reaction.

Mix on ice.

- Vector: (pBIG1a, 1b, etc) 50-100ng
- Inserts: 5 fold molar excess for inserts smaller than the vector and a 2 fold excess for inserts of similar or larger size (4kb or bigger).
- Water to 5µL

Add this to a 5µL aliquot of 2x assembly mix.

- Incubate at 50°C for 15-60 minutes. (15 minutes is fine for two fragment assembly, but more fragments require longer).
- Transform 2µL of assembly into 20µL of cells that come with the kit (any highly competent cells for cloning purposes should be fine), plate onto amp+spec LB plates and grow over night at 37°C. I like to do a control using an equivalent amount of vector without assembly mix to give an idea of how many background colonies there may be. For the first assembly reaction you ever do, you could also use the assembly control provided with the Gibson assembly kit to ensure your assembly mix works as expected.

Screening by colony PCR

Hopefully your transformation plates have ~30 or more colonies.

I've designed a set of 4 primers (S1A-D) for screening with colony PCR whether assembly has occurred. When all primers are used for a PCR reaction, this results in three products if assembly has been successful, but only one of size 350bp if no insert has assembled (figure 4). Partial assemblies are possible, so all three bands need to be present. I find screening at least 5 colonies of assembly reaction is advisable, if you assemble more than 3 genes, or the genes are variable sizes, I would recommend you screen as many as practical (using strip PCR tubes or plates means this can be high through-put).

I use the KAPPA robust PCR kit, because it always work for me:

https://www.n-genetics.com/file/KAPA2G_Robust_TDS_KR0379_v7.16.pdf

- Leave transformation plates until colonies are a reasonable size.

- Circle and number the colonies you wish to screen and pick them into 80µL of sterile water with a pipette tip, being careful not to scope up any agar as this can lead to false positive. Don't pick up too much colony, as the PCR won't work. The water solution should not be cloudy.

<i>PCR mix :</i>	<i>1x</i>	<i>10x</i>
dNTPs (10 mM)	0.2 µL	2 µL
primer mix (10 µM of each)	0.5 µL	5 µL
5x Buffer B	2 µL	20 µL
Kappa robust enzyme	0.1 µL	1 µL
water	6.2 µL	62 µL

- Mix on ice, aliquot 9µL per tube, then add 1µL of colony water.
- Use annealing temperature of 58°C and an extension time of 30s, for 30-35 cycles.
- Run PCR product on a 2% gel and check band pattern.
- Set up over-night cultures for mini-preps of 3 or 4 colonies which resulted in correct three bands.

Diagnostic digest:

To determine if all genes have been assembled, perform a diagnostic digest with Swal. This will cut between all the expression GECs so you should get a DNA band for each of your genes, where each will be the size of your gene plus about 500bp. (I make a plasmid map of what I expect to make using the program APE (A Plasmid Editor), a get a predicted band pattern for digestion with Swal: <http://biologylabs.utah.edu/jorgensen/wayned/ape/>).

Mini prep DNA	4µL
10x buffer	1µL
water	4.5µL
Swal	0.5µL

- Incubate for a couple of hours at room temperature and run on 1% gel. Shorter digests are possible, but if digest is incomplete, the resultant DNA gel can be very hard to interpret (you will get extra bands).

If you have very low efficiency, with mainly partially assembled products, see if you can find a pattern in what is going wrong. If one band is frequently missing, try add more of that during the assembly reaction. You can also try change the order of your genes, which seems to help if genes are very different sizes.

If band pattern is correct, proceed to pBIG2 assembly. At this point you can also send your pBIG1 constructs for sequencing. I typically use biGBac to make 5-10 constructs at once and my genes are huge, so I tend to test two or three different constructs for expression and confirm the sequence by mass spec and sequencing after a construct shows good expression. This saves sequencing several 20kb constructs that may not even express.

Assembly of pBIG2 constructs

- Digest pBIG1s and pBIG2s with PmeI. Digesting the pBIG1 assemblies with PmeI also acts as a screening step, so ensure you check the band size is what you expect.

Mini prep (5-10µg)	20-30µL
10x cutsmart buffer	4 µL
PmeI enzyme	1-2µL
Water	Up to 50µL

- Incubate at 37°C for a few hours for pBIG1 vectors, and overnight for pBIG2 vectors.
- Run 1% gel, check that band formed by PmeI digest is the expected size for the genes you have assembled, using the APE program then gel extract using the zymo kit, eluting in 10µL of water.
- Perform assembly as above, this time plating on Amp+Cam plates.

Screening by colony PCR

- Screen as above using second set (S2A-D) of screening primers. These will result in 424, 322 & 680 bp products if assembly has occurred or 800 bp if no insert is present.

Diagnostic digest

- Grow over nights for mini-preps, and perform a diagnostic digest with PacI. This cuts out each fragment assembled in to pBIG2 vectors. You can also check using a Swal digest, which will cut out each of your gene expression cassettes. This can result in a lot of bands, so I normal do diagnostic digests with both enzymes separately to check everything is there.

Expression in Insect cells

After you have the biGBac constructs, standard insect cells protocols apply. Transform your pLIB, pBIG1 or pBIG2 constructs into DH10emBacY cells, and purify the bacmid. Make P1 virus stock by transfecting the Bacmid into Sf9 cells, after 3 days amplify the virus by adding transfection supernanant to make a P2 virus stock for expression. Be aware that if you have a large construct (>20kb) the P2 may take longer than usual. You should also take extra care of your cells to make sure they don't over grow between passages, as this can greatly effect protein expression especially for large complexes.

Discussion

Construct Design

Construct design is really important with biGBac cloning. I encourage you to think about all the ways you might want to use your constructs before you start.

Things to consider are do you just want the whole complex, or are there sub-complexes you may also want to investigate? Has this complex been purified before so you know where to put the tags? Chances are you will have to change this as you optimise your purification or perform some other down stream experiment. To prevent having to go through multi-fragment assembly again and again, I suggest you think about making every pBIG1 vector a usable sub-complex and make any tags used removable at a DNA level using flanking unique restriction enzyme sites. You can also put a unique restriction site where you may want to add a tag, such that adding the tag is as easy as digesting the pBIG1 or pBIG2 assembled vector and using Gibson assembly to add the tag sequence. Although I haven't rigorously tested different types of tags, some repetitive sequence tags seem to make the cloning slightly more challenging. Hence, there is logic in making constructs without affinity tags and just restriction sites, then add a range of different tags in a systematic way.

It may also be the case you want make a whole gene removable and exchangeable, for example if you want to make a series of mutations in a catalytic site or similar. To do this you simply have to add a unique restriction enzyme site in the Cas primers between the pLIB homology region and the variable region.

As the restriction sites cannot be present in any of your genes, or in the pBIG1s or pBIG2s, you won't have a lot of choice, so this requires thought before ordering the genes. I used XbaI, NdeI, SpeI, KsaI and XhoI for these purposes as I had these enzymes at hand.

The largest source of false positives is from incomplete digestion of vectors. This is a particular problem for pBIG1 vectors as SmaI enzyme which linearises it cuts at room temperature and is inactivated at higher temperatures. Weissmann do partially address this problem, by performing a long digest, adding the enzyme in two separate rounds. An alternative way to linearise the vector is by PCR (using primers alpha_rev and omega_for in table 2), then remove the vector template by DpnI digestion (will remove any DNA purified from *E. coli*) and gel extraction. The yield for this typically is not great and does come with a small risk of backbone mutation, but using this method I get no vector background. Doing backbone PCR also enables further modification of the system.

The overlapping regions (α , A, β , B, γ , etc) were chosen by Weissmann based on computational optimisation to prevent secondary structure and lack cross complementarity, but they never actually test whether their algorithm for choosing the overlaps performed better than a random piece of DNA ~25bp long or test whether longer pieces of DNA might be more efficient. Given Gibson *et al* (Gibson *et al* 2010) were able to assemble a 1.08 Mega base bacterial genome, assembling 10 fragments at a time, the sequence of the overlaps should not be so sensitive and 5 fragments should not produce such low efficiency. Companies such as DNA technology at Synthetic genomes incorporated, for which Daniel G. Gibson is the Vice president, suggest overlaps of ≥ 40 bp if you want to assemble 5 genes, especially if those genes are over 8 kb. Hence, 5-fragment biGBac assembly could be more efficient if the overlaps were extended. As the pBIG1 assembly is mediated by overlaps introduced by PCR, these overlaps could

easily be extended by just lengthening the PCR primer. The vector overlaps could also be lengthened by linearising the vector using PCR.

Alternatively, as the biGBac system is a set of Gibson assembly homology regions that allow DNA fragment assembly and contain restriction enzyme sites to release DNA fragments and screen for correct assemblies, this concept could be further extended to make another pBIG vector. Additional level of concatenation is possible by incorporating an additional blunt cutting enzyme site flanking Gibson assembly overlaps in the vector of the pBIG1s and adding another screening restriction enzyme site (Figure 5). This must be coupled with minor changes in the Cas primers to ensure Gibson assembly overlaps are still complementary, but by using this approach 27 genes could be assembled with only three fragments being assembled at a time. This would add two days to the 6 day protocol, but is likely to have a higher efficiency.

Conclusion

The biGBac system is a great modular way of making a large amount of expression constructs for complex molecular assemblies. By fully understanding all of the features, I hope you can see how this could be adapted and extended.

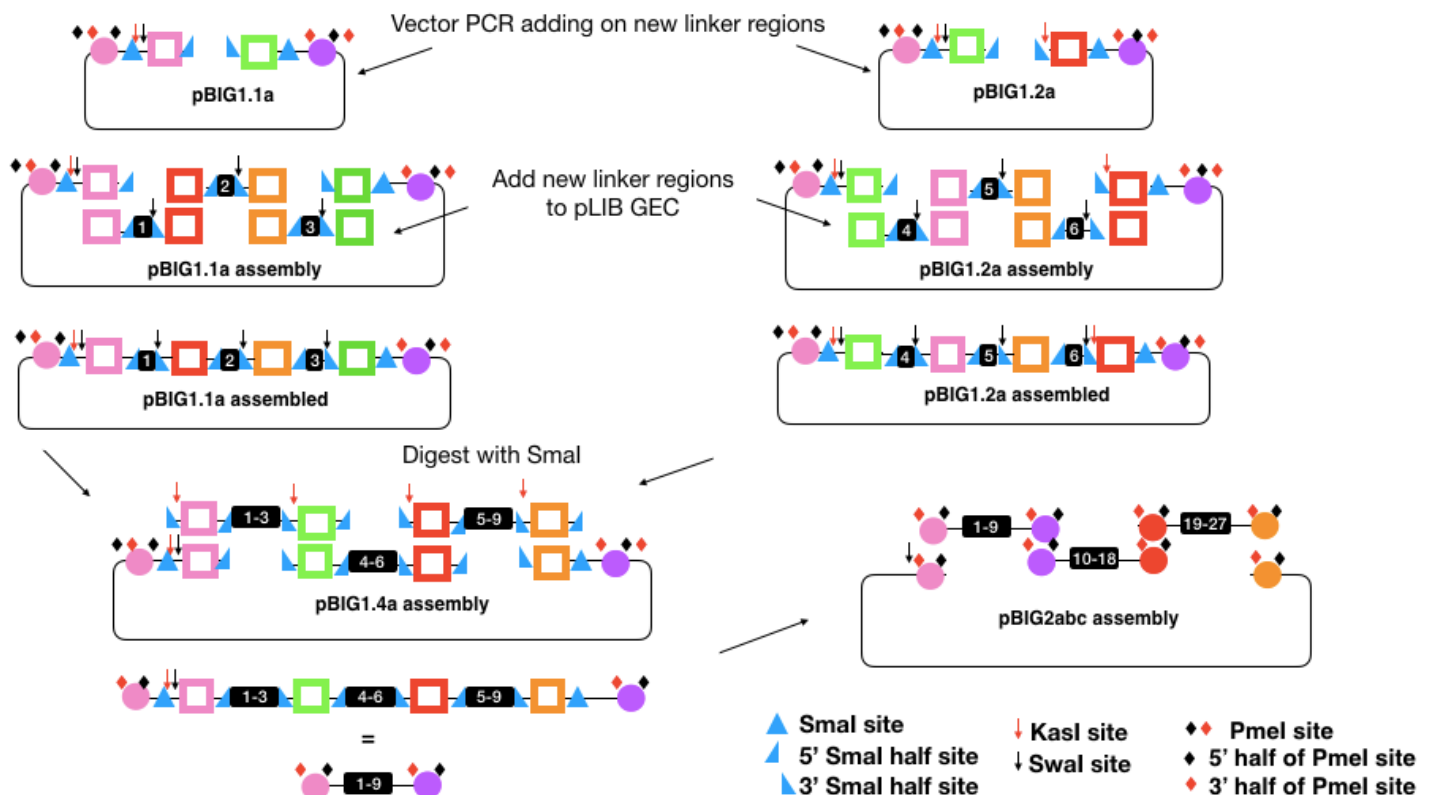


Figure 5: Introducing a third assembly step could be achieved by modifying the overlapping regions in pBIG1 assembly to include a new blunt cutting enzyme site (eg SmaI) for cassette removal and a new enzyme site for screening (KasI). These overlaps would have to be added to both GECs from pLIB and the pBIG1 vectors, which could be achieved by PCR. After the first round of assembly, digestion with SmaI would release the cassette allowing an additional step on assembly into another vector based off pBIG1. As the PmeI sites would still be present in this modified pBIG1, DNA fragments from PmeI digest could be assembled in pBIG2s.

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Table 1: Vectors in the biGBac system

Vector	Description	Antibiotic resistance
pLib	Stores individual genes, with promoter and terminator.	Amp, Genta
pBIG1a	Stores up to 5 genes expression cassettes	Amp, Genta, Spec
pBIG1b		Amp, Genta, Spec
pBIG1c		Amp, Genta, Spec
pBIG1d		Amp, Genta, Spec
pBIG1e		Amp, Genta, Spec
pBIG2ab	Stores pBIG1a and 1b cassettes	Amp, Genta, Cam
pBIG2abc	Stores pBIG1a, 1b and 1c cassettes	Amp, Genta, Cam
pBIG2abcd	Stores pBIG1a, 1b, 1c and 1d cassettes	Amp, Genta, Cam
pBIG2abcde	Stores pBIG1a, 1b, 1c, 1d and 1e cassettes	Amp, Genta, Cam

Table 2: Primers used in cloning and screening

Name	Sequence	Notes
CasI_for	aacgctctatggtctaaagatttAAATCGACCTACTCCG GAATATTAATAGATC	Adds α or A overlap at 5'
CasI_rev	aaacgtgcaatagatccagtttATTAAATGGTTATGA TAGTTATTGCTCAGCG	Adds β or C overlap at 3'
CasII_for	aaactggatactattgcacggttAAATCGACCTACTCCG GAATATTAATAGATC	Adds β or C overlap at 5'
CasII_rev	aaacatcaggcatcattaggttATTAAATGGTTATGA TAGTTATTGCTCAGCG	Adds γ or D overlap at 3'
CasIII_for	aaacctaagatgcctgatgttAAATCGACCTACTCCG GAATATTAATAGATC	Adds γ or D overlap at 5'
CasIII_rev	aaactaagctatgtgaaccgttATTAAATGGTTATGA TAGTTATTGCTCAGCG	Adds δ or E overlap at 3'
CasIV_for	aaacggttcacatagcttagttAAATCGACCTACTCCG GAATATTAATAGATC	Adds δ or E overlap at 5'
CasIV_rev	aaaccaagtcaatgtcagtggttATTAAATGGTTATGA TAGTTATTGCTCAGCG	Adds ϵ or F overlap at 3'
CasV_for	aaacactgacattgactgggttAAATCGACCTACTCCG GAATATTAATAGATC	Adds ϵ or F overlap at 5'
Cas ω _rev	aaccccgattgagatatagattATTAAATGGTTATGA TAGTTATTGCTCAGCG	Adds ω or B overlap at 3'
S1A	CGAGGCTAACTGAAACACGG	pBIG1 screen A: Binds pBIG1 vectors between A/B/.. and alpha linker
S1B	CCCGTAGTGGCTATGGCAG	pBIG1 screen B: Binds pBIG1 vectors between omega and B/C/... linkers
S1C	GCGAGATGGTTATCATTTTAATTATCTCC	pBIG1 screen C: Binds pLIB vector at poly H promoter
S1D	CCTCCACACCTCCCCCTG	pBIG1 screen D: Binds pLIB vector at terminator
S2A	GACGGTTTGTACATGGAG	pBIG2 screen A: binds both pBIG1 and 2 after linker site
S2B	CAGTTGCGTGAGCGCATACGC	pBIG2 screen B: binds both pBIG1 and 2 before loxP site
S2C	CGTCGATACCCACCGAG	pBIG2 screen C: binds pBIG1 vector between omega and B/C/...
S2D	CACTTGACAGAGTGCCAGCC	pBIG2 screen D: binds pBIG1 vector between A/B/... and alpha
Alpha_rev	AAATCTTTAGACCATAGAGCGTTCACCTGACAG AGTGC	
Omega_for	AAATCTATATCTCAATCGGGGTACGTCGATAC CCCAC	pBIG2 screen D: binds pBIG1 vector between A/B/... and alpha